

SOLUTIONS

illuminati - 2019

Solutions to Advanced Mathematics Assignment-1A | Calculus

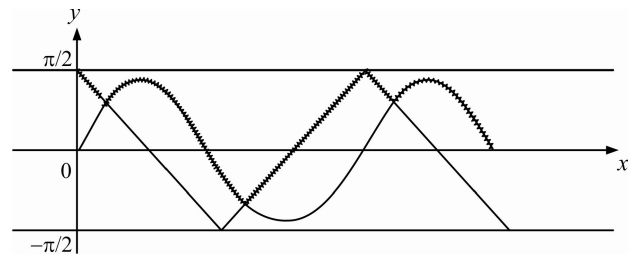
Section 1
Single Correct Answer Type

1.(B) We know that $\lim_{x \rightarrow 0^+} x^x = 1 \Rightarrow \lim_{x \rightarrow 0} (x^x - 1) = 0$

$$\begin{aligned} \text{So, } \lim_{x \rightarrow 0^+} \frac{e^{(x^x-1)} - x^x}{\left((x^x)^x - 1\right)^2} &= \lim_{x \rightarrow 0^+} \frac{\left(1 + (x^x - 1) + \frac{(x^x - 1)^2}{2!} + \frac{(x^x - 1)^3}{3!} + \dots\right) - x^x}{(x^x - 1)^2 (x^x + 1)^2} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{(x^x - 1)^2}{2!} + \frac{(x^x - 1)^3}{3!} + \dots}{(x^x - 1)^2 (x^x + 1)^2} = \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) = \frac{1}{8} \end{aligned}$$

2.(B) $f(x) = \max\{\sin x, \sin^{-1}(\cos x)\}$

Clearly, $f(x)$ is continuous but non-differentiable at infinite points.



3.(C) (A) As, $\sin x < x \forall x > 0 \Rightarrow \sin x - x = 0$ has no root in $\left(0, \frac{\pi}{2}\right)$

(B) $\tan x > x \forall x \in \left(0, \frac{\pi}{2}\right) \Rightarrow \tan x - x = 0$ has no root in $\left(0, \frac{\pi}{2}\right)$

(C) The statement is true by extreme value theorem (Property of continuous function in closed interval)

(D) The function $g(x)$ is not given continuous in $[3, 5]$ so the given statement is not always true.

4.(B) $(f(x))^2 - (e^x + e^{x^2})f(x) + e^x \cdot e^{x^2} \leq 0 \Rightarrow (f(x) - e^x)(f(x) - e^{x^2}) \leq 0$

$$e^{x^2} \leq f(x) \leq e^x; \forall x \in (0, 1) \Rightarrow e^x \leq f(x) \leq e^{x^2}; \forall x \in (1, \infty)$$

$\therefore \lim_{x \rightarrow 1} (e^x) = \lim_{x \rightarrow 1} (e^{x^2}) = e$ Hence, $\lim_{x \rightarrow 1} f(x) = e$ by sandwich theorem.

5.(B) Given, $\Delta_r = \begin{vmatrix} r(r-1) & 2(r+2) & 2015 \\ r & 1 & -6 \\ r^2 & 4r & 0 \end{vmatrix}$

Take r common from I column and III row

$$\Delta_r = r^2 \begin{vmatrix} (r-1) & 2(r+2) & 2015 \\ 1 & 1 & -6 \\ 1 & 4 & 0 \end{vmatrix} \Rightarrow \Delta_r = \begin{vmatrix} r^2(r-1) & 2r^2(r+2) & 2015r^2 \\ 1 & 1 & -6 \\ 1 & 4 & 0 \end{vmatrix}$$

$$\text{Now, } \sum_{r=1}^n \Delta_r = \begin{vmatrix} \sum r^3 - \sum r^2 & 2\sum r^3 + 4\sum r^2 & 2015\sum r^2 \\ 1 & 1 & -6 \\ 1 & 4 & 0 \end{vmatrix} \quad \text{Hence, } \lim_{n \rightarrow \infty} \left(\frac{\sum_{r=1}^n \Delta_r}{n^4} \right) = \begin{vmatrix} \frac{1}{4} & \frac{1}{2} & 0 \\ 1 & 1 & -6 \\ 1 & 4 & 0 \end{vmatrix} = 6 \left(1 - \frac{1}{2} \right) = 3$$

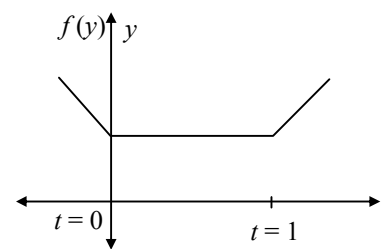
$$\begin{aligned} 6.(B) \quad L &= \lim_{x \rightarrow 0^+} \left(\frac{\cot^{-1}(\ln(x))}{\pi} \right)^{-\ln x} (1^\infty) = e^{\lim_{x \rightarrow 0^+} \left(\frac{\cot^{-1}(\ln(x)) - \pi}{\pi} \right) (-\ln x)} = e^{\lim_{x \rightarrow 0^+} \frac{\cot^{-1}(-\ln x)}{\pi} (\ln x)} \\ &= e^{-\lim_{x \rightarrow 0^+} \frac{\tan^{-1}\left(-\frac{1}{\ln x}\right)}{\left(-\frac{1}{\ln x}\right)\pi}} = e^{-\frac{1}{\pi}} \quad \therefore \text{L.H.L.} = f(0^-) = e^{\left(-\frac{x}{\pi}, \frac{1}{x}\right)} = e^{-\frac{1}{\pi}} \Rightarrow f(0^+) = f(0^-) = e^{-\frac{1}{\pi}} = k \end{aligned}$$

$$7.(B) \quad \text{Given } f(t) = \begin{cases} 1-2t & , \quad t < 0 \\ 1 & , \quad 0 \leq t \leq 1 \\ 2t-1 & , \quad t > 1 \end{cases} \Rightarrow g(x) = \begin{cases} f(x-1) & , \quad 0 \leq x \leq 1 \\ 3-x & , \quad 1 < x \leq 2 \end{cases}$$

$$\Rightarrow g(x) = \begin{cases} 3-2x & , \quad 0 \leq x \leq 1 \\ 3-x & , \quad 1 < x \leq 2 \end{cases}$$

$\therefore$ The function $g(x)$ is discontinuous at $x = 1$, as $\lim_{x \rightarrow 1} g(x)$ does not exist.

So, $g(x)$ is non-differentiable at $x = 1$



$$8.(C) \quad \because \frac{\pi}{3} < A < \frac{\pi}{2} \Rightarrow \frac{\sqrt{3}}{2} < \sin A < 1 \text{ and } \frac{3\sqrt{3}}{8} < \sin^3 A < 1 \Rightarrow 0 < \sin A - \sin^3 A < 1 \text{ and } \sin A + \sin^3 A > 1$$

$$\text{Now } f(x) = \begin{cases} ax^2 & , \quad x \in Q \\ 5x-b & , \quad x \notin Q \end{cases} \Rightarrow f(x) \text{ is continuous at } x = 2 \text{ and } 3.$$

$$\text{So, } ax^2 = 5x - b \text{ should have roots 2 and 3} \Rightarrow ax^2 - 5x + b = 0$$

$$\text{So, } \frac{5}{a} = 5 \Rightarrow a = 1 \text{ and } \frac{b}{a} = 6 \Rightarrow b = 6 \Rightarrow b - a = 5$$

$$9.(C) \quad \therefore f(x) \text{ is differentiable in } (0, \infty).$$

Hence, $\lim_{x \rightarrow \infty} f(x)$ must exist and is finite.

$\therefore y = f(x)$ must have a horizontal asymptote as $x \rightarrow \infty$ then only $\lim_{x \rightarrow \infty} f(x)$ will exist.

If $f(x)$ has an inclined asymptotes as $y = x - \frac{1}{x}$ then $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$.

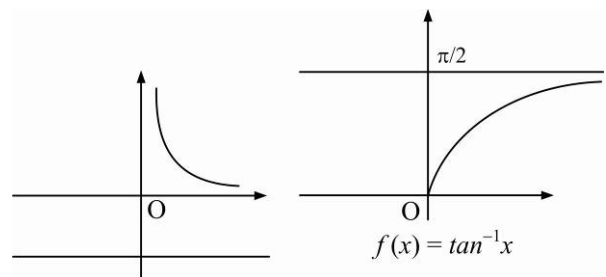
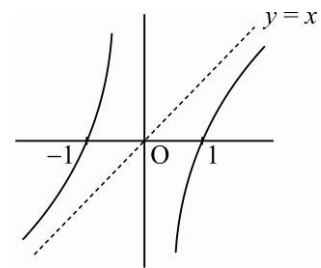
$\therefore f(x)$ has a horizontal asymptote

$$\text{Hence, } \lim_{x \rightarrow \infty} f'(x) \rightarrow \infty$$

$\Rightarrow$ (C) (also see figure for $f(x) = \tan^{-1} x$)
e.g., Take the example given

(i) Let $f(x) = x \sin \frac{1}{x}$ which is differentiable in $(0, \infty)$

$$f'(x) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}$$



$$f(x) + f'(x) = \underbrace{\left(x \sin \frac{1}{x}\right)}_{\lim_{x \rightarrow \infty}} + \underbrace{\left(x \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}\right)}_{\lim_{x \rightarrow \infty} \rightarrow 0}$$

Hence, $\lim_{x \rightarrow \infty} f(x) = L$ and $\lim_{x \rightarrow \infty} f'(x) = 0$

(ii) $f(x) = \tan^{-1} x$ in $(0, \infty)$

10.(D) Using L'Hospital rule 2 times, we get limit = 4

11.(D) Here $f(x) = 1 - x \Rightarrow f'(x) = -1 \Rightarrow f(x) = 1 - x = f^{-1}(x)$, i.e., inverse of itself.

12.(D) Here $g(h(x)) = x \Rightarrow g'(h(x))h'(x) = 1$

$$\Rightarrow \frac{1}{1 + (h(x))^3} \cdot h'(x) = 1 \Rightarrow h'(x) = 1 + (h(x))^3$$

$$\Rightarrow h''(x) = 3(h(x))^2 h'(x) \Rightarrow h''(x) = 3h^2(x)(1 + h^3(x))$$

13.(B) Given, $\int_{-2T}^{a+5T} f(x) dx = 19$

$$\Rightarrow \int_{-2T}^{5T} f(x) dx + \int_{5T}^{a+5T} f(x) dx = 19$$

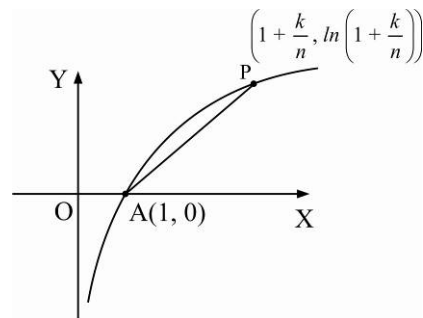
$$\Rightarrow 7 \int_0^T f(x) dx + \int_0^a f(x) dx = 19$$

$$\Rightarrow \int_0^a f(x) dx = 19 - 7 \times 2 = 19 - 14 = 5$$

14.(A) $\lim_{n \rightarrow \infty} \frac{n^\beta}{n^\alpha} \frac{\sum_{r=1}^n r^\alpha}{\sum_{r=1}^n r^\beta} = \frac{\int_0^1 x^\alpha dx}{\int_0^1 x^\beta dx} = \frac{\beta+1}{\alpha+1}$

16.(B) $L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^4 \left(\left(\frac{k}{n} \right)^2 + \ln^2 \left(1 + \frac{k}{n} \right) \right)$
 $= \int_0^1 (x^2 + \ln^2(1+x)) dx = \frac{1}{3} + 2\ln^2 \frac{2}{e}$

15.(C)



17.(C) $\int_0^1 (1-x^7)^{1/4} dx - \int_0^1 (1-x^4)^{1/7} dx$

$$y = f(x) = (1-x^7)^{1/4} \Rightarrow y^4 = (1-x^7) \Rightarrow x = (1-y^4)^{1/7}$$

Hence functions are inverse of each other.

$$I = \int_0^1 f(x) dx - \int_0^1 g(y) dy = \int_0^1 f(x) dx - \int_1^0 x f'(x) dx$$

Hence, $I = \int_0^1 (f(x) + x f'(x)) dx = x f(x) \Big|_0^1 = f(1) = 0$

18.(C)

19.(C) $Y - y = m(X - x) dx$

Now, $\left(\frac{y-0}{x-1} \right) m = -1$

$$y \frac{dy}{dx} = 1 - x$$

Integrating, $\frac{y^2}{2} = x - \frac{x^2}{2} + C$

$$x^2 + y^2 - 2x = C \quad \dots \dots (i)$$

This is the equation of a circle with centre (1, 0).

$\therefore 2x + 3y - 15 = 0$ is tangent at (i).

$\therefore$ Perpendicular from (1, 0) on the line = r

$$\left| \frac{2-15}{\sqrt{13}} \right| = \sqrt{1+C}$$

$$C+1=13 \Rightarrow C=12$$

Hence, the curve is $x^2 + y^2 - 2x = 12$

20.(B) $\sin x \frac{dy}{dx} + y \cos x = 1$

$$\frac{dy}{dx} + y \cot x = \operatorname{cosec} x$$

$$\text{I.F.} = e^{\int \cot x dx} = e^{\ln(\sin x)} = \sin x$$

$$y \sin x = \int \operatorname{cosec} x \cdot \sin x dx$$

$$y \sin x = x + C$$

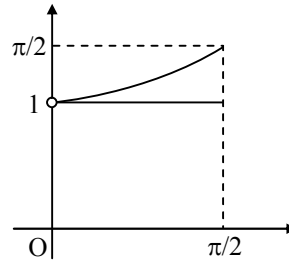
If $x = 0$, y is finite.

$\therefore C = 0$

$$y = x(\operatorname{cosec} x) = \frac{x}{\sin x}$$

Now, $I < \frac{\pi^2}{4}$ and $I > \frac{\pi}{2}$

Hence, $\frac{\pi}{2} < I < \frac{\pi^2}{4} \Rightarrow (A)$



21.(C) $x \int_0^x g(t) dt + \int_0^x (1-t)g(t) dt = x^4 + x^2$

Differentiate w.r.t. x : $xg(x) + \int_0^x g(t) dt + (1-x)g(x) = 4x^3 + 2x$

Again differentiate w.r.t. x : $g'(x) + g(x) = 12x^2 + 2$

Now, $\int_0^1 \frac{12}{g'(x) + g(x) + 10} dx = \int_0^1 \frac{dx}{x^2 + 1} = \left[\tan^{-1} x \right]_0^1 = \frac{\pi}{4}$

22.(A) $dy + xy dx = x dx$

$$\frac{dy}{dx} + xy = x \Rightarrow \text{Integrating factor} = e^{x^2/2} \quad \therefore y e^{x^2/2} = \int x e^{x^2/2} dx = e^{x^2/2} + C$$

Hence, $y = 1 + C e^{-x^2/2}$

$$y(0) = 2 \Rightarrow C = 1$$

Hence, $y(x) = 1 + e^{-x^2/2}$

$$A = \int_0^\infty \left(e^{-x^2/2} + 1 \right) - 1 dx = \int_0^\infty e^{-x^2/2} dx$$

23.(A) I.F. = $e^{\int \frac{g'(x)}{g(x)} dx} = e^{\ln g(x)} = g(x)$

$$y g(x) = \frac{1}{2} \int \frac{2g(x)g'(x)}{1+g^2(x)} dx = \frac{1}{2} \ln[1+g^2(x)] + C$$

Here, $C = 1 - \frac{\ln 2}{2}$

$$y(e)g(e) = \frac{1}{2} \ln(1+2e-1) + 1 - \frac{\ln 2}{2} = \frac{3}{2}$$

24.(D) $\frac{f(x)}{1+x^2} = 1 + \int_0^x \frac{f^2(t)dt}{1+t^2}$ [Note : $f(0) = 1$]

Differentiate both sides w.r.t. x

$$\frac{(1+x^2)f'(x) - f(x)(2x)}{(1+x^2)^2} = \frac{f^2(x)}{(1+x^2)}$$

$$\frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right)y = y^2$$

$$\frac{1}{y^2} \frac{dy}{dx} - \left(\frac{2x}{1+x^2}\right)\frac{1}{y} = 1$$

Let $\frac{-1}{y} = t \Rightarrow \frac{dt}{dx} + \left(\frac{2x}{1+x^2}\right)t = 1$

Solving the above L.D.E.

$$f(x) = \frac{-3(1+x^2)}{x^3 + 3x - 3} \Rightarrow f(-2) = \frac{-3(5)}{-8-6-3} = \frac{15}{17}$$

25.(B)

26.(A) $z = \sqrt{x^2 + 4} + \sqrt{y-4} \quad (y > 4)$

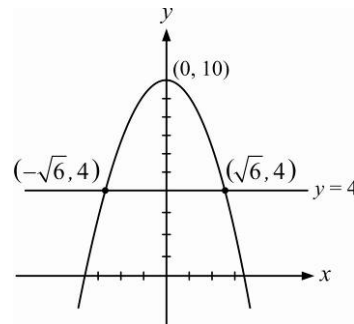
$$\left| \sqrt{x^2 + 4} + i\sqrt{y-4} \right| = \sqrt{10}$$

$$x^2 + 4 + y - 4 = 10 \Rightarrow x^2 + y = 10$$

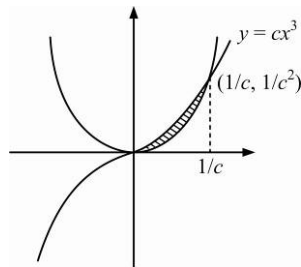
$$y = 10 - x^2$$

But $4 \leq y \leq 10$

$$A = 2 \int_0^{\sqrt{6}} \left[(10 - x^2) - 4 \right] dx = 2 \left[6x - \frac{x^3}{3} \right]_0^{\sqrt{6}} = 8\sqrt{6}$$



27.(C)



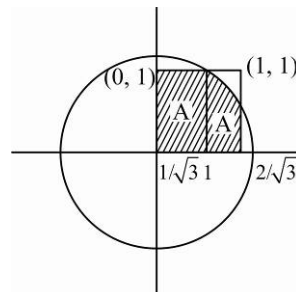
28.(C) $A = \frac{1}{\sqrt{3}} \times 1 + A_1$

$$A_1 = \int_{1/\sqrt{3}}^1 \left(\sqrt{\left(\frac{4}{3}\right) - x^2} \right) dx$$

Put $x = \frac{2}{\sqrt{3}} \sin \theta \Rightarrow dx = \frac{2}{\sqrt{3}} \cos \theta d\theta$

$$= \int_{\pi/6}^{\pi/3} \frac{2}{\sqrt{3}} \cos \theta \cdot \frac{2}{\sqrt{3}} \cos \theta d\theta = \frac{2}{3} \int_{\pi/6}^{\pi/3} (\cos 2\theta + 1) d\theta = \frac{2}{3} \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\pi/6}^{\pi/3}$$

$$= \frac{2}{3} \left[\frac{1}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\pi}{3} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\pi}{6} \right] = \frac{2}{3} \cdot \frac{\pi}{6} = \frac{\pi}{9}$$



$$A = \frac{1}{\sqrt{3}} + \frac{\pi}{9} = \frac{3\sqrt{3} + \pi}{9}$$

$$a = 3, b = 1$$

$$a + b = 4$$

29.(D) $f'(x) = 2x + 1$

$$\Rightarrow f(x) = x^2 + x + c$$

Given $f(1) = 3 \quad \therefore c = 1$

$$f(x) = x^2 + x + 1$$

$$\frac{dy}{dx} = (1+x)^y \left(\frac{y}{1+x} + \ln(x+1)y' \right) + \frac{1}{\sqrt{1-\sin^4 x}} (2 \sin x \cos x)$$

At $x = 0, y = 1$

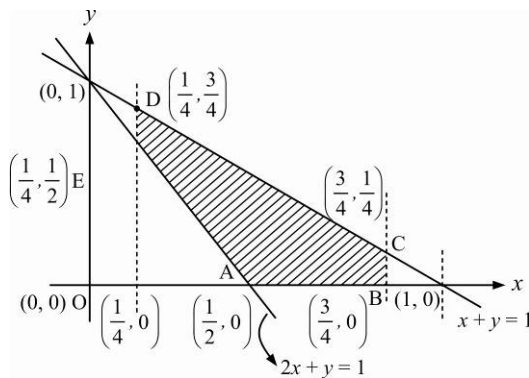
$$m_T = 1$$

$\therefore$ Normal $y - 1 = -1(x - 0)$

$$x + y = 1$$

$$\text{Required area} = \int_{-2}^0 \left((1-x) - (x^2 + x + 1) \right) dx = - \int_{-2}^0 (2x + x^2) dx = - \left(x^2 + \frac{x^3}{3} \right)_{-2}^0 = - \left[0 - \left(4 - \frac{8}{3} \right) \right] = \frac{4}{3}$$

30.(B) We have, $x + y \leq 1, 2x + y \geq 1$ and $(4x - 3)(4x - 1) \leq 0$



$\therefore$ From above graph, option (B) is correct.

Section 2

Link Comprehension Type

31-32. 31.(A) 32.(C)

$$\frac{1}{f(x+y)} f'(x+y) = \frac{1}{f(x)} f'(x) \ln f(y)$$

Putting $x = 0$: $\frac{f'(x)}{f(y)} = \frac{f'(0)}{f(0)} \ln f(y)$

$$\Rightarrow \frac{f'(y)}{f(y) \ln f(y)} = 1 \quad \{ \because f(0) = e, f'(0) = e \}$$

$$\Rightarrow \ln(\ln f(y)) = y + c \Rightarrow c = 0 \quad \Rightarrow f(y) = e^{e^y} \Rightarrow f(x) = e^{e^x}$$

Now $g(x) = \ln(\ln(x))$ and $g'(x) = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}$

$$g''(x) = -\frac{x \cdot \frac{1}{x} + \ln x}{(x \ln x)^2} = -\frac{1 + \ln x}{(x \ln x)^2} \Rightarrow g''(e) = -\frac{1+1}{e^2} = -\frac{2}{e^2}$$

$$\ln(\ln f(x)) = g^2(f(x) - 3g(f(x))) - 5 \Rightarrow x = x^2 - 3x - 5$$

$$x^2 - 4x - 5 = 0 \Rightarrow (x-5)(x+1) = 0 \Rightarrow x = -1, 5$$

33-35. 33.(D) 34.(C) 35.(B)

$$F(x) = \int_0^x f(t) dt \Rightarrow F'(x) = f(x)$$

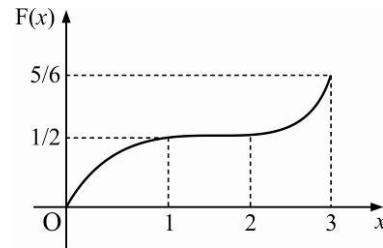
$$\text{For } 0 \leq x \leq 1 : F(x) = \int_0^x (1-t) dt = \left[t - \frac{t^2}{2} \right]_0^x$$

$$F(x) = x - \frac{x^2}{2} \quad \text{and} \quad F(1) = \frac{1}{2}$$

$$\text{For } 1 < x \leq 2 : F(x) = \int_0^1 (1-t) dt + \int_1^x 0 dt = \frac{1}{2} + 0 \Rightarrow F(x) \text{ is constant}$$

$$\text{For } 2 < x \leq 3 : F(x) = \int_0^1 f(t) dt + \int_1^2 f(t) dt + \int_2^x f(t) dt = \frac{1}{2} + 0 + \int_2^x (t-2)^2 dt = \frac{1}{2} + \frac{(x-2)^3}{3}$$

$$\therefore F(x) = \begin{cases} x - \frac{x^2}{2} & \text{if } 0 \leq x \leq 1 \\ \frac{1}{2} & \text{if } 1 < x \leq 2 \\ \frac{1}{2} + \frac{(x-2)^3}{3} & \text{if } 2 < x \leq 3 \end{cases}$$



(ii) Range of $F(x)$ from the graph $\left[0, \frac{5}{6}\right]$

$$(iii) \text{ Area} = \int_0^1 F(x) dx + \frac{1}{2} + \int_2^3 F(x) dx$$

$$= \int_0^1 \left(x - \frac{x^2}{2} \right) dx + \frac{1}{2} + \int_2^3 \left(\frac{1}{2} + \frac{(x-2)^3}{3} \right) dx = \left[\frac{x^2}{2} - \frac{x^3}{6} \right]_0^1 + \frac{1}{2} + \left[\frac{x}{2} + \frac{(x-2)^4}{12} \right]_2^3$$

$$= \frac{1}{2} + \left(\frac{1}{2} - \frac{1}{6} \right) + \left[\left(\frac{3}{2} + \frac{1}{12} \right) - (1) \right] = \frac{1}{2} + \frac{1}{3} + \frac{7}{12} = \frac{6+4+7}{12} = \frac{17}{12}$$

36-38 36.(D) 37.(C) 38.(B)

$$g(x) = x - k \text{ where } k = \int_0^1 f(t) dt \quad \dots (i)$$

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt = \frac{x^3}{2} + 1 - x \int_0^x (t-k) dt = \frac{x^3}{2} + 1 - x \left(\frac{t^2}{2} - kt \right) \Big|_0^x = \frac{x^3}{2} + 1 - x \left(\frac{x^2}{2} - kx \right) = 1 + kx^2$$

$$f(x) = 1 + kx^2 \quad \dots (ii)$$

$$\text{From (i) : } k = \int_0^1 (1 + kt^2) dt = t + \frac{kt^3}{3} \Big|_0^1 = 1 + \frac{k}{3} \Rightarrow \frac{2k}{3} = 1 \Rightarrow k = \frac{3}{2}$$

$$\therefore g(x) = x - \frac{x}{3} \Rightarrow g \text{ is linear with slope } 1 \text{ \& cuts the } y\text{-axis at } \frac{3}{2} \Rightarrow (B)$$

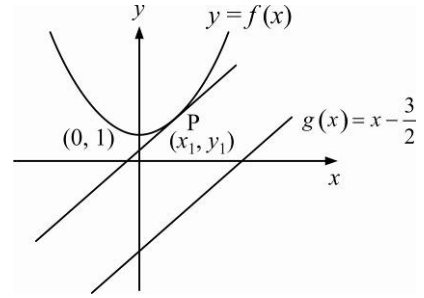
$$\text{Also } f(x) = 1 + \frac{3x^2}{2} \text{ which is a quadratic polynomial } \Rightarrow (D)$$

$$\text{Now } f'(x) \Big|_P = 3x_1 = 1$$

$$\Rightarrow x_1 = \frac{1}{3}$$

$$y_1 = 1 + \frac{3}{2} \cdot \frac{1}{9} = 1 + \frac{1}{6} = \frac{7}{6}$$

$$\text{Hence, } P\left(\frac{1}{3}, \frac{7}{6}\right)$$

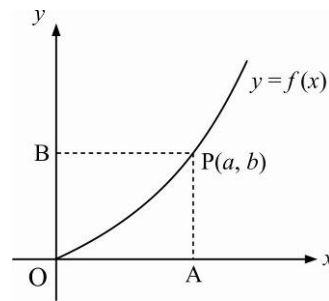


$$\text{Perpendicular distance from P on the line } y = x - \frac{3}{2} \text{ is } d = \left| \frac{\frac{1}{3} - \frac{7}{6} - \frac{3}{2}}{\sqrt{2}} \right| = \left| \frac{-\frac{5}{6} - \frac{3}{2}}{\sqrt{2}} \right| = \left| \frac{-\frac{7}{3}}{\sqrt{2}} \right| = \frac{7}{3\sqrt{2}}$$

39-41. 39.(B) 40.(B) 41.(A)

$$\frac{\text{Area of } OPBO}{\text{Area of } OAPO} = \frac{2}{1} \Rightarrow \frac{ab - \int_0^a f(x) dx}{\int_0^a f(x) dx} = \frac{2}{1}$$

$$\Rightarrow ab = 3 \int_0^a f(x) dx$$



$$\therefore \text{Locus of } (a, b) \text{ is curve } \therefore xy = 3 \int_0^x f(x) dx$$

$$\text{On differentiating both sides, we get : } x \frac{dy}{dx} + y = 3y \Rightarrow x \frac{dy}{dx} = 2y$$

$$\Rightarrow \int \frac{dy}{y} = 2 \int \frac{dx}{x} \Rightarrow \ln y = 2 \ln x + \ln c \Rightarrow y = cx^2$$

$$\frac{dy}{y} = \frac{2x}{x} \Rightarrow y = cx^2$$

$$\text{Now } (2, 4) \text{ is on it } \Rightarrow c = 1 \Rightarrow \text{curve is } y = x^2$$

$$(i) \text{ Any tangent to } y^2 = 8x \text{ is } y = mx + \frac{2}{m}$$

$$\text{Solving (i) with } y = x^2, \text{ we get : } x^2 - mx - \frac{2}{m} = 0$$

Put discriminant = 0 (for tangency)

$$\Rightarrow m^2 = \frac{-8}{m} \Rightarrow m = -2$$

$$\therefore \text{Equation of common tangent is } y = -2x - 1 \Rightarrow 2x + y + 1 = 0$$

(ii) Line through (3, 0)

$$y - 0 = m(x - 3)$$

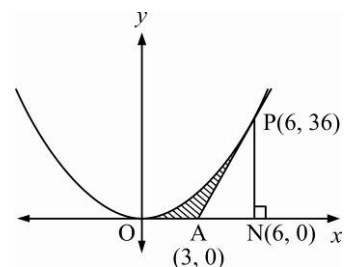
Solving it with $y = x^2$

$$mx - 3m = x^2 \Rightarrow x^2 - mx + 3m = 0$$

Put discriminant = 0 (for tangency)

$$\text{So, } m^2 - 12m = 0 \Rightarrow m = 0 \text{ or } m = 12$$

$$\text{Equation of tangent is } y = 12x - 36 \text{ and } y = 0.$$



42-43. 42.(A) 43.(C)

$$f_n(x) = \sum_{n=1}^n \frac{\sin^2 x}{\cos^2\left(\frac{x}{2}\right) - \cos^2\left(\frac{2n+1}{2}\right)x} = \sin x \sum_{n=1}^n \frac{\sin\{(n+1)x - nx\}}{\sin(n+1)x \cdot \sin(nx)}$$

$$= \sin x \sum_{n=1}^n \cot(nx) - \cot(n+1)x = \sin x (\cot x - \cot(n+1)x) = \frac{\sin(nx)}{\sin(n+1)x}$$

$$g_n(x) = f_1(x) \cdot f_2(x) \cdot f_3(x) \cdot \dots \cdot f_n(x) = \frac{\sin(x)}{\sin(n+1)x}$$

$$I_n = \int_x^\pi \frac{\sin(nx)}{\sin x} dx = \begin{cases} 0 & \text{if } n \text{ is even} \\ \pi & \text{if } n \text{ is odd} \end{cases}$$

$$I_{n+2} = \int_0^\pi \frac{\sin(n+2)x}{\sin x} dx$$

$$I_{n+2} - I_n = \int_0^\pi \frac{\sin(n+2)x - \sin nx}{\sin x} dx = 2 \int_0^\pi \cos(n+1)x dx = 0$$

$$I_{n+2} = I_n$$

$$I_1 = \pi \quad \text{and} \quad I_0 = 0$$

$$\Rightarrow I_0 = I_2 = I_4 = \dots = 0 \quad \text{and} \quad I_1 = I_3 = I_5 = \dots = \pi$$

$$\sum_{k=1}^{100} I_k = (I_1 + I_2 + I_3 + \dots + I_{100}) = (I_2 + I_4 + \dots + I_{100}) + (I_1 + I_3 + \dots + I_{99}) = 50\pi$$

$$\lim_{x \rightarrow 0} \int_0^x \frac{9dt}{xf_9(t)g_9(t)} = \lim_{x \rightarrow 0} \frac{9}{x} \int_0^x \frac{\sin^2(10)t}{\sin(9t)\sin t} dt \left(\frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{9 \sin^2 10x}{\sin 9x \sin x} = 100$$

44-47. (1) $f(x+y) = e^x f(y) + e^y f(x) \dots (i)$

$$x = y = 0 \Rightarrow f(0) = 2f(0) \rightarrow f(0) = 0$$

Differentiating (i) w.r.t. x partially $f'(x+y) = e^x f(y) + e^y f'(x) \dots (ii)$

Differentiating (i) w.r.t. y partially $f'(x+y) = e^x f'(y) + e^y f(x) \dots (iii)$

$$(ii)-(iii) \Rightarrow \frac{f'(x) - f(x)}{e^x} = \frac{f'(y) - f(y)}{e^y} \rightarrow f'(x) - f(x) = Ae^x, \text{ where } A \text{ is a constant.}$$

It is linear d.e. with integrating factor e^{-x} .

$$\therefore f(x)e^{-x} = \int Ae^x \cdot e^{-x} dx = Ax + B$$

$$f(0) = 0 \rightarrow B = 0$$

$$\therefore f(x) = Axe^x$$

$$f'(x) = A(x+1)e^x$$

$$f'(0) = 1 \rightarrow A = 1$$

$$\therefore f(x) = xe^x, f'(x) = (x+1)e^x$$

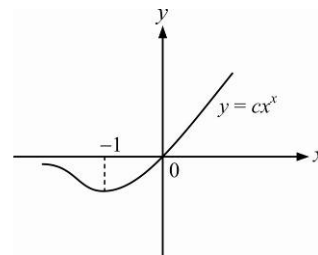
$$f'(x) = 0 \rightarrow x = -1$$

$$f''(x) = (x+2)e^x, f''(-1) = e^{-1} > 0 \Rightarrow f(x) \text{ has minimum at } x = -1$$

(2) $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ The range is $[f(-1), f(\infty)) = \left[-\frac{1}{e}, \infty\right)$

(3) $\lim_{x \rightarrow -\infty} \frac{x}{e^{-x}} = \lim_{x \rightarrow -\infty} \frac{1}{-e^{-x}} = 0$ by L.H. rule

(4) The area = $\int_{-\infty}^0 -f(x) dx = \int_{-\infty}^0 -xe^x dx = -(x-1)e^x \Big|_{-\infty}^0 = 1$



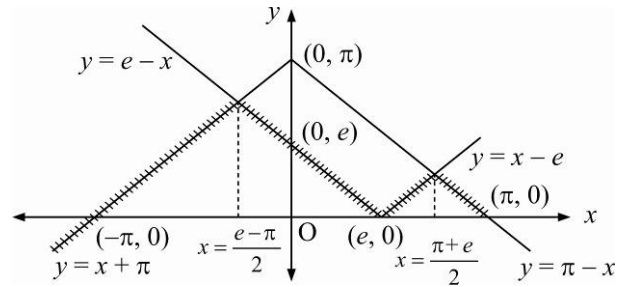
Section 3

Multiple Correct Answer Type

48.(CD) Clearly, $f(x)$ is continuous everywhere but non-derivative at 3 points viz. $x = \frac{e-\pi}{2}, e, \frac{\pi+e}{2}$

Now verify alternatives from graph.

Also, range of $f(x)$ is $\left(-\infty, \frac{\pi+e}{2}\right]$.



49.(ABCD)
$$\lim_{x \rightarrow 0^+} \frac{\frac{p \cos x}{e^{1/x}} + x}{\frac{1 + \sin x}{e^{1/x}} + q \cos x} = \frac{0}{0+q} = 0 \Rightarrow q \in \mathbb{R} - \{0\}$$

$$\lim_{x \rightarrow 0} \frac{p \cos x + x e^{1/x}}{1 + \sin x + q \cos x e^{1/x}} = p = 0 \Rightarrow p = 0, q \in \mathbb{R} - \{0\}$$

50.(BD)
$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{P(1/x^3)}{e^{1/x^4}} = \lim_{t \rightarrow \infty} \frac{P(t^3)}{e^{t^4}} = 0 \quad (\text{As we know } \lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0, n \in \mathbb{N}) \Rightarrow \lim_{x \rightarrow 0} f(x) = f(0) = 0$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{P(1/h^3)}{h e^{1/h^4}} = \lim_{t \rightarrow \infty} \frac{t P(t^3)}{e^{t^4}} = 0$$

51.(AD) Given, $f: (0, \infty) \rightarrow (-\infty, \infty)$ be defined as $f(x) = e^x + \ln x$

Also, $f'(x) = e^x + \frac{1}{x} > 0 \forall x \in (0, \infty)$

Also, $\lim_{x \rightarrow +\infty} f(x) = \infty$ and $\lim_{x \rightarrow 0^+} f(x) = -\infty \Rightarrow f(x)$ is a bijective mapping.

As $f(1) = e \Rightarrow g(e) = 1$

Also, $g'(e) = \frac{1}{f'(1)}$

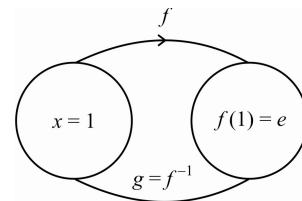
$$f'(x) = e^x + \frac{1}{x} \Rightarrow f'(1) = (e+1)$$

$$f''(x) = e^x - \frac{1}{x^2} \Rightarrow f''(1) = (e-1)$$

So, $g'(e) = \frac{1}{e+1}$

Also, $g''(e) = \frac{-f''(1)}{(f'(1))^3} \quad \left(\text{As, } g''(x) = \frac{-f''(x)}{(f'(x))^3} \right)$

$$= \frac{-(e-1)}{(e+1)^3} = \frac{1-e}{(e+1)^3} \text{ now verify the choices.}$$



52.(BD)
$$f(x) = \int \frac{x}{2} d\left(1 - \frac{1}{x^2}\right) = \int \frac{x}{2} \cdot \frac{2}{x^3} dx = \int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

$$f(2) = -\frac{1}{2} + C = \frac{1}{2} \Rightarrow C = 1 \quad \therefore f(x) = \frac{x-1}{2}$$

$$g_1(x) = f(x) = \frac{x-1}{2}; \quad g_2(x) = f(f(x)) = \frac{\frac{x-1}{2} - 1}{\frac{x-1}{2}} = \frac{-1}{x-1} = \frac{1}{1-x};$$

$$g_3(x) = f(f(f(x))) = \frac{1}{1 - \left(\frac{x-1}{x}\right)} = x$$

$$g_4(x) = g_1(x) = \frac{x-1}{x}; g_5(x) = g_2(x) = \frac{1}{1-x}; g_6(x) = g_3(x) = x \text{ and so on}$$

$$\therefore g_{(3n-2)}(x) = \frac{x-1}{x}, g_{(3n-1)}(x) = \frac{1}{1-x}, g_{3n}(x) = x$$

$$\frac{d}{dx} \left(g_{(3n-2)}(x) \right) = \frac{d}{dx} \left(1 - \frac{1}{x} \right) = \frac{1}{x^2} \quad \therefore \text{Option (A) is wrong}$$

$$\frac{d}{dx} \left(g_{3n} x^{(n)} \right) = \frac{d}{dx} (x) = 1 \quad \therefore \text{Option (B) is wrong}$$

$$\lim_{x \rightarrow 1} \frac{x + x^2 + \dots + x^{100} + x - 101}{x - 1}$$

Applying L'Hospital's rule

$$\lim_{x \rightarrow 1} \frac{1 + 2x + \dots + 100x^{99} + 1}{1} = 1 + 2 + \dots + 100 + 1 = 5051 \quad \therefore \text{Option (C) is wrong}$$

$$y = g_{10}(x) = g_{(3n-1)}(x) = \frac{1}{1-x} \quad \therefore g'_{(3n-1)}(x) = \frac{1}{(1-x)^2}$$

$$g'_{(3n-1)}(x) \Big|_{x=\frac{1}{2}} = 4 \quad \therefore \text{Option (D) is correct}$$

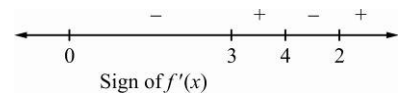
53.(BCD) We have $f'(x) = \frac{3^x}{1+x^2} > 0 \forall x > 0$

$$\therefore f'(x) = \frac{3^x}{1+x^2} > \frac{1}{1+x^2} > 0 \forall x > 0 \Rightarrow \int_1^x f'(x) dx > \int_1^x \frac{1}{1+x^2} dx \Rightarrow f(x) > \tan^{-1} x - \tan^{-1} 1$$

So, $f(x) + \frac{\pi}{4} > \tan^{-1} x$, for all $x > 0$

Note that in (D) equality holds at $x = 1$ and hence (D) is false.

54.(ACD) We have $f(x) = \int_0^x e^{-t^2} (t-5)(t^2 - 7t + 12) dt$



$$\therefore f'(x) = e^{-x^2} (x-5)(x^2 - 7x + 12) = e^{-x^2} (x-3)(x-4)(x-5)$$

Now, verify options (A) and (B).

As, $f'(3) = 0 = f'(4) = f'(5)$

$\therefore$ Consider $y = f'(x)$, on $x \in [3, 4]$

As Rolle's theorem is applicable. So, $\exists$ some $c_1 \in (3, 4)$ such that $f''(c_1) = 0$

Similarly, consider $y = f'(x)$, on $x \in [4, 5]$. As Rolle's theorem is applicable

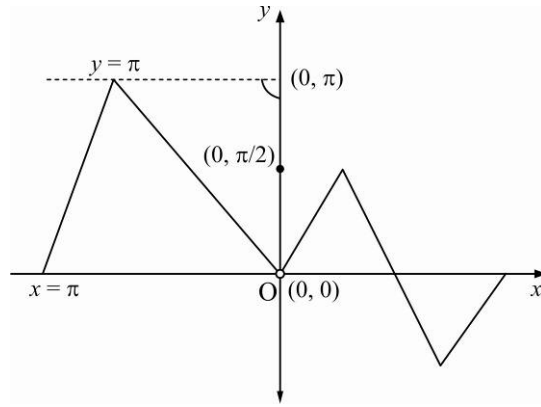
So, $\exists$ some $c_2 \in (4, 5)$ such that $f''(c_2) = 0$

$\Rightarrow$ There exists atleast two $c_1, c_2 \in (0, \infty)$ such that $f''(c_1) = 0$ and $f''(c_2) = 0 \Rightarrow$ (C) is correct.

Now consider $y = f''(x)$, $x \in [c_1, c_2]$. As Rolle's theorem is applicable

So $f'''(c) = 0$ for some $c \in (0, \infty) \Rightarrow$ Option (D) is correct.

55.(AC)



Now verify alternative.

56.(ABD) $f(-1) = -f(1)$ & $f(0) = 0$ as is odd.

Consider the interval $x \in [-1, 0]$:

$$\text{Slope of chord joining } (-1, f(-1)) \text{ \& } (0, f(0)) = \frac{f(-1) - f(0)}{-1 - 0} = \frac{-f(1) - 0}{-1 - 0} = f(1)$$

So applying LMVT we get : $\exists a \in (-1, 0)$ such that $f'(a) = f(1)$

Similarly considering the interval $x \in [0, 1]$ and applying LMVT we get :

$$\exists b \in (0, 1) \text{ such that } f'(b) = f(1)$$

Now, consider the function $g(x) = f^n(x)$, $x \in [-1, 1]$

Consider the interval $x \in [-1, 0]$

$$\begin{aligned} \text{Slope of chord joining } (-1, g(-1)) \text{ \& } (0, g(0)) &= \frac{g(-1) - g(0)}{-1 - 0} \\ &= \frac{f^n(-1) - f^n(0)}{-1} = \frac{(-1)^n f^n(1) - 0}{-1} = (-1)^{n-1} f^n(1) = \begin{cases} f^n(1) & \text{if } n \text{ is odd} \\ -f^n(1) & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

So, option (C) is correct only when n is odd.

$$\text{But in the interval } x \in [0, 1] : \text{ Slope of chord joining } (0, g(0)) \text{ \& } (1, g(1)) = \frac{g(1) - g(0)}{1 - 0} = f^n(1)$$

$$\text{So, LMVT} \Rightarrow \exists \beta \in (0, 1) \text{ such that } g'(\beta) = f^n(1)$$

57.(ABCD) Given, $f(f(x)) = -x + 1$

Replacing $x \rightarrow f(x)$

$$\Rightarrow f(f(f(x))) = -f(x) + 1 \Rightarrow f(1-x) = -f(x) + 1 \Rightarrow f(x) + f(1-x) = 1 \dots (i)$$

$$\text{Now, } J = \int_0^1 f(x) dx = \int_0^1 f(1-x) dx \quad [\text{Using } (a-x) \text{ property}]$$

$$\Rightarrow 2J = \int_0^1 (f(x) + f(1-x)) dx \Rightarrow J = \frac{1}{2}$$

$$\text{Also, from equation (i), we get : } f''(x) + f''(1-x) = 0 \Rightarrow f''\left(\frac{1}{2}\right) = 0$$

$$\text{So, } f''\left(\frac{1}{2}\right) = f''\left(\frac{4}{5}\right) \quad \text{Hence, by Rolle's theorem, } f'''(x) = 0 \Rightarrow x \in \left(\frac{1}{2}, \frac{4}{5}\right)$$

$$\text{Also, } f'(x) = f'(1-x) \Rightarrow f'\left(\frac{1}{4}\right) = f'\left(\frac{3}{4}\right)$$

$$\text{So, by Rolle's theorem } f''(x) = 0 \Rightarrow x \in \left(\frac{1}{4}, \frac{3}{4}\right)$$

58.(AC) $f(x) = e^x \sec x - \sqrt{2} \cos x + x$

$$f'(x) = e^x \sec x \tan x + \sec x \cdot e^x + \sqrt{2} \sin x + 1 = e^x \sec x (\tan x + 1) + (\sqrt{2} \sin x + 1)$$

Clearly, $f(x)$ is minimum at $x = \frac{-\pi}{4}$

$$f(x)|_{\min} = f\left(\frac{-\pi}{4}\right) = \left(\sqrt{2}e^{-\pi/4} - 1 - \frac{\pi}{4}\right)$$

$$f''(x) > 0 \quad \forall x \in \left[\frac{-\pi}{3}, \frac{\pi}{3}\right] = f'(x) \text{ is increasing} \quad \Rightarrow \quad f'\left(\frac{\pi}{3}\right) \geq f'(x) \quad \forall x \in \left[\frac{-\pi}{3}, \frac{\pi}{3}\right]$$

59.(BCD)
$$f(x) = \frac{1 - \cos\{x\}}{x^2(x^3 + ax^2 + bx + c)^2}$$

$$x^3 + ax^2 + bx + c = (x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6 \quad \therefore \quad a + b + c = -1$$

$$f(x) = \frac{1 - \cos\{x\}}{x^2((x-1)(x-2)(x-3))^2}$$

Now, $\ell = \frac{1}{8}, m = \frac{1}{8}$ and $n = \frac{1}{72} \quad \therefore \quad \ell + m + n = \frac{19}{72}$

60.(ABD) Given, $f'(t) = e^t (\cos^2 t - \sin 2t)$

$\therefore$ Integrate both sides with respect to t , we get : $f(t) = e^t \cos^2 t + C$

But, $f(0) = 1 \Rightarrow C = 0$ So, $f(t) = e^t \cos^2 t$

Clearly, [As, $-\infty < t < 0$ so $0 < e^t < 1$ and $0 \leq \cos^2 t \leq 1 \Rightarrow 0 \leq e^t \cos^2 t \leq 1 \Rightarrow 0 \leq e^t \cos^2 t \leq 1 \Rightarrow f$ is bounded in $(-\infty, 0)$]

Note that f is neither odd nor even function.

Also, $f(t) = e^t \Rightarrow e^t \cos^2 t = e^t \Rightarrow \cos^2 t = 1 \quad \therefore \quad t = 0, \pi, 2\pi \in [0, 2\pi]$. So, 3 solutions exist.

Also, $\lim_{t \rightarrow 0} (f(t))^{1/t} \text{ (} 1^\infty \text{ form)} = e^L$, where $L = e^{\lim_{t \rightarrow 0} \left(\frac{e^t \cos^2 t - 1}{t} \right) \left(\frac{0}{0} \text{ form} \right)}$

$$\left[\text{where : } \lim_{t \rightarrow 0} \frac{e^t (\cos^2 t - \sin 2t)}{1} = e^t \right] \text{ (Using L'Hospital rule)}$$

$$\Rightarrow \quad \ell = e^1 = e$$

61.(ABC) We have, $\int_0^x f(t) dt + x \int_0^x f(t) dt - \int_0^x t \cdot f(t) dt = (-1 + e^{-x})$

Differentiate both sides with respect to x , we get :

$$f(x) + \int_0^x f(t) dt = -e^{-x} \quad \dots \text{(i)}$$

Put $x = 0$, we get : $f(0) = -1$

Again, differentiate both sides of equation (i) with respect to x , we get : $e^x (f'(x) + f(x)) = 1$

$\therefore$ On integrating both sides with respect to x , we get : $e^x \cdot f(x) = x + c$

As, $f(0) = -1 \quad \Rightarrow \quad -1 = 0 + c \quad \Rightarrow \quad c = -1$

$\therefore \quad f(x) = (x-1)e^{-x}$

Now verify alternatives.

62.(AC)

63.(ABC) $f(x) = \frac{\sin x}{x}$

64.(ABD) $\frac{dy}{dx} + y = f(x)$

I.F. = $e^x \Rightarrow ye^x = \int e^x f(x) dx + C$

Now if $0 \leq x \leq 2$ then $ye^x = \int e^x e^{-x} dx + C \Rightarrow ye^x = x + C$

$x = 0, y(0) = 1, C = 1$

$\therefore ye^x = x + 1 \quad \dots (i)$

$y = \frac{x+1}{e^x}; y(1) = \frac{2}{e} \Rightarrow y' = \frac{e^x - (x+1)e^x}{e^{2x}}$

$y'(1) = \frac{e-2e}{e^2} = \frac{-e}{e^2} = -\frac{1}{e} \Rightarrow (B) \text{ is correct.}$

if $x > 2$

$ye^x = \int e^{x-2} dx \Rightarrow ye^x = e^{x-2} + C$

$y = e^{-2} + Ce^{-x}$

As y is continuous

$\therefore \lim_{x \rightarrow 2} \frac{x+1}{e^x} = \lim_{x \rightarrow 2} (e^{-2} + Ce^{-x})$

$3e^{-2} = e^{-2} + Ce^{-2} \Rightarrow C = 2$

$\therefore$ for $x > 2$

$y = e^{-2} + 2e^{-x};$ hence $y(3) = 2e^{-3} + e^{-2} = e^{-2}(2e^{-1} + 1)$

$y' = -2e^{-x}$

$y'(3) = -2e^{-3} \Rightarrow (D) \text{ is correct.}$

65.(AC) Given, $\frac{dy}{dx} - 3y \cot x = \sin 2x, y\left(\frac{\pi}{2}\right) = 2$ [Linear differential equation]

$\therefore$ I.F. = $e^{-\int 3 \cot x dx} = e^{-3 \ln(\sin x)} = \frac{1}{\sin^3 x}$

$\therefore$ General solution is $y\left(\frac{1}{\sin^3 x}\right) = \int \frac{2 \sin x \cdot \cos x}{\sin^3 x} dx + C = 2 \int \operatorname{cosec} x \cot x dx + C$

$\Rightarrow \frac{y}{\sin^3 x} = -2 \operatorname{cosec} x + C$

As, $y\left(x = \frac{\pi}{2}\right) = 2 \Rightarrow \frac{2}{1^3} = -2 + C \Rightarrow C = 4 \quad \therefore y = (4 \sin^3 x - 2 \sin^2 x)$

(A) $y\left(\frac{\pi}{6}\right) = 4\left(\frac{1}{2}\right)^3 - 2\left(\frac{1}{2}\right)^2 = \frac{1}{2} - \frac{1}{2} = 0$

(B) $y'(x) = (12 \sin^2 x - 4 \sin x \cos x)$ So, $y'\left(\frac{\pi}{3}\right) = \left(\frac{12 \times 3}{4} \times \frac{1}{2} - 4 \times \frac{\sqrt{3}}{2} \times \frac{1}{2}\right) = \frac{9}{2} - \frac{2\sqrt{3}}{2} = \left(\frac{9-2\sqrt{3}}{2}\right)$

(C) As, $y'(x) = 4 \sin x \cos x (3 \sin x - 1) = 2 \sin 2x (3 \sin x - 1)$

$\therefore y(x)$ increase in $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

(D) $\int_{-\pi/2}^{\pi/2} \underbrace{(4 \sin^3 x)}_{\text{Odd function}} - \underbrace{2 \sin^2 x}_{\text{Even function}} dx = 0 - 4 \int_0^{\pi/2} \sin^2 x dx = -4 \left(\frac{\pi}{4}\right) = -\pi$

66.(ABCD)

According to the given information,

$$-\int_0^r f(x)dx = \frac{2}{5} \quad \dots(i)$$

$$\int_r^s f(x)dx = 2 \quad \dots(ii)$$

$$-\int_s^6 f(x)dx = 12 \quad \dots(iii)$$

$$m = \int_s^0 f(x)dx = -\int_0^s f(x)dx = -\int_0^r f(x)dx - \int_r^s f(x)dx = \frac{2}{5} - 2 = \frac{-8}{5} = -1.6$$

$$p = \int_r^6 f(x)dx = \int_r^s f(x)dx + \int_s^6 f(x)dx = 2 - 12 = -10$$

$$q = \left| \int_0^6 f(x)dx \right| = \left| \int_0^r f(x)dx + \int_r^s f(x)dx + \int_s^6 f(x)dx \right| = \left| \frac{-2}{5} + 2 - 12 \right| = \frac{52}{5} = 10.4$$

$$n = \int_r^6 |f(x)| dx = \int_0^r |f(x)| dx + \int_r^s |f(x)| dx + \int_s^6 |f(x)| dx$$

$$= -\int_0^r f(x)dx + \int_r^s f(x)dx - \int_s^6 f(x)dx = \frac{2}{5} + 2 + 12 = \frac{72}{5} = 14.4$$

$$m = -1.6 ; p = -10 ; q = 10.4 ; n = 14.4$$

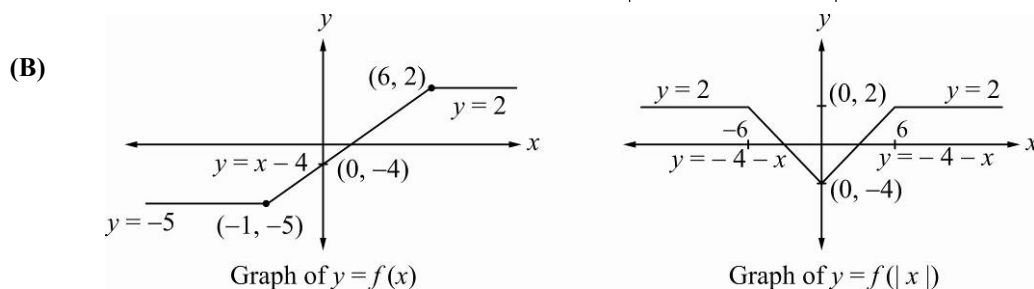
Section 4

Match Matrix Type

67. [A-s] [B-p] [C-r] [D-q]

(A) Clearly, $|x^2 - 9| + |x^2 - 4| = 5 = |(x^2 - 9) - (x^2 - 4)|$

So, $(x^2 - 9)(x^2 - 4) \leq 0$ (As $|a| + |b| = 5 = |(x^2 - 9) - (x^2 - 4)| \Rightarrow x \in [-3, -2] \cup [2, 3]$)



Clearly, $R_{f(|x|)} = [-4, 2]$

$\therefore$ Number of integers in the range of function $f(|x|) = 7$

Now, clearly $\lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)} = \lim_{x \rightarrow 7} \frac{([x]-7)([x]-6)}{(x-7)(x-6)} = 0 \Rightarrow (P)$

(C)
$$L = \lim_{x \rightarrow 0} \frac{a \left(x - \frac{x^3}{3!} + \dots \right) + bx \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \right) + 3x^2}{\left(\frac{\sin x - x}{x^3} - \frac{(x - \tan x)}{x^3} \right) x^3}$$

$$L = \lim_{x \rightarrow 0} \frac{x(a+b) + x^2(b+3) + x^3\left(\frac{b}{2} - \frac{a}{6}\right) + \dots}{x^3} \Rightarrow L = 6\left(\frac{b}{2} - \frac{a}{6}\right)$$

$$\Rightarrow a+b=0 \text{ and } b+3=0 \Rightarrow b=-3. \text{ Also } a=3$$

$$L = 6\left(-\frac{3}{2} - \frac{3}{6}\right) = -12. \text{ Hence, } \left(\frac{b-L}{a}\right) = \frac{-3-(-12)}{3} = \frac{9}{3} = 3 \Rightarrow (R)$$

(D) We have $1 - \frac{1}{k+1}C_2 = 1 - \frac{2}{(k+1)k} = \frac{k^2+k-2}{k(k+1)} = \frac{(k+2)(k-1)}{(k+1)k}$

$$\text{Hence, } P_n = \prod_{k=2}^n \frac{k-1}{k} \cdot \prod_{k=2}^n \frac{k+2}{k+1} = \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \dots \cdot \frac{n-1}{n}\right) \left(\frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \dots \cdot \frac{n+2}{n+1}\right)$$

$$\text{Hence, } P_n = \frac{1}{n} \left(\frac{n+2}{3}\right) = \frac{n+2}{3n}$$

$$\lim_{n \rightarrow \infty} P_n = \frac{1}{3} \Rightarrow a=1, b=3, \text{ hence } (b-a)=2 \Rightarrow (Q).$$

68. [A-p, r] [B-p, q, r] [C-r] [D-p, q]

(A) $\ell n(f(x)) = \ell n(x-1) + \ell n(x-2) + \dots + \ell n(x-n)$

$$f'(x) = f(x) \left[\frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n} \right]$$

$$f'(x) = (x-2)(x-3) \dots (x-n) + (x-1)(x-3) \dots (x-n) + \dots + (x-1)(x-2) \dots (x-(n-1))$$

$$f'(n) = (n-1)(n-2)(n-3) \dots 3 \cdot 2 \cdot 1 \text{ (all other factors except the last vanishes when } x=n)$$

$$5040 = (n-1)!$$

$$7! = (n-1)! \Rightarrow n=8 \Rightarrow (P), (R)$$

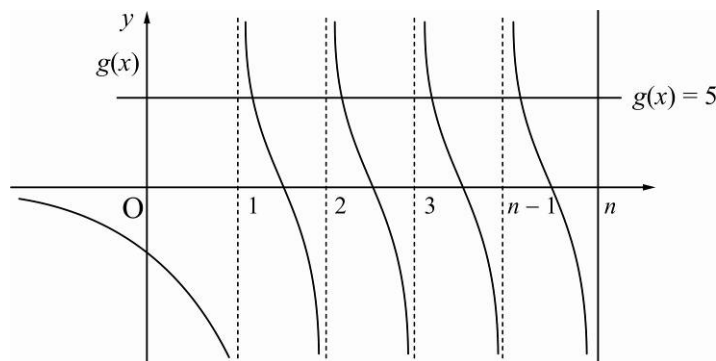
(B) $g(x) = \frac{f'(x)}{f(x)} = \frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n} + C$ Hence, $g'(x) < 0$

(C) $g(x)$ is discontinuous at n points

$$\Rightarrow g(x) = 5 \text{ has } n \text{ roots} \Rightarrow g(x) = \frac{f'(x)}{f(x)} \text{ or } g(x) = \frac{f'(x)}{f(x)} + C \therefore n=8 \Rightarrow (R)$$

(D) Number of roots $f'(x) = 0$ is $n-1$ (Use L.M.V.T.)

$$\Rightarrow (n-1)(n-5)^2 = n-1 \Rightarrow (n-1)(n-6)(n-4) = 0 \Rightarrow (P), (Q)$$



69. [A-r] [B-s] [C-t] [D-q]

(A) A, B, C are the 3 critical points of $y = f(x)$

$f''(x) = 0$ for $x = 2$ and fails to exist at $x = 0$.

(B) $x = \frac{1}{4}$ and 2. Make a quadratic in $\log_2 x$ and get the answer

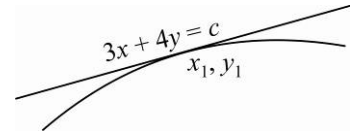
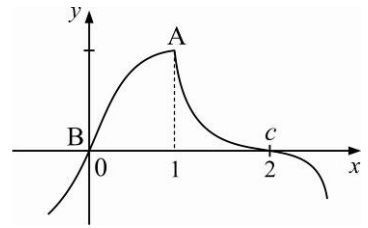
$$(C) \quad \frac{dy}{dx} = -1 + 2x_1^3 = -\frac{3}{4} \Rightarrow x_1 = \frac{1}{2}$$

$$\Rightarrow \quad \frac{1}{32} = \frac{1}{2} + y_1 \text{ or } y_1 = -\frac{15}{32} \Rightarrow c = -\frac{3}{8}$$

(D) $f'(x) = 2x^3 - 3x + 1$ this is always positive in $(1, 2)$.

$\therefore$ Increasing in $[1, 2]$

$\therefore f(2)$ will be the greatest value.



70. [A-s] [B-t] [C-q] [D-r]

$$(A) \quad 3 \int_0^1 \sin(2\pi + 3\pi x) dx = 3 \int_0^1 \sin 3\pi x dx = -\frac{3}{3\pi} [\cos 3\pi x]_0^1 = -\frac{1}{\pi} [(-1) - (1)] = \frac{2}{\pi} \Rightarrow (S)$$

$$(B) \quad I = \int_0^\infty [x] e^{-x} dx = \int_0^1 [x] e^{-x} dx + \int_1^2 [x] e^{-x} dx + \int_2^3 [x] e^{-x} dx + \dots$$

$$= 0 + 1 \int_1^2 e^{-x} dx + 2 \int_2^3 e^{-x} dx + 3 \int_3^4 e^{-x} dx + \dots$$

$$= (e^{-1} - e^{-2}) + 2(e^{-2} - e^{-3}) + 3(e^{-3} - e^{-4}) + \dots \infty = e^{-1} + e^{-2} + e^{-3} + e^{-4} + \dots \infty = \frac{e^{-1}}{1 - e^{-1}} = \frac{1}{e - 1} \Rightarrow (T)$$

$$(C) \quad \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{2i}{n} \right)$$

$$\sum_{i=1}^n \frac{\ln 2 + \ln \left(\frac{i}{n} \right)}{n} = \int_0^1 (\ln 2 + \ln x) dx$$

$$= [x \ln 2 + x \ln x - x]_0^1 = (\ln 2 - 1) - 0 \quad (\text{As } \lim_{x \rightarrow 0} x \ln x = 0)$$

$$= \ln 2 - 1 \Rightarrow k = \frac{2}{e} \Rightarrow (Q)$$

$$(D) \quad \ell = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{i}{n} \sin \left(\frac{i^2}{n^2} \right) \pi = \int_0^1 x \sin(\pi x^2) dx = \frac{1}{\pi} \Rightarrow (R)$$

71.(A) [A-p] [B-r] [C-p]

$$(A) \quad \int_0^1 x f'''(2x) dx = \frac{x f''(2x)}{2} \Big|_0^1 - \frac{1}{2} \int_0^1 f''(2x) dx$$

$$\frac{f''(2)}{2} - \left[\frac{f'(2x)}{4} \Big|_0^1 \right] = 2 \Rightarrow (P)$$

(B) $f(x)$ simplifies to $x^2 + x + 1$

$$\therefore \quad I = \int_0^1 (x^2 + x + 1) dx = \frac{1}{3} + \frac{1}{2} + 1 = \frac{11}{6} \Rightarrow (R)$$

$$\begin{aligned}
 \text{(C)} \quad I &= \int_0^{2\pi} \left| (x^2 - \pi^2) - \sin^2 x \right| dx \\
 I &= \int_0^{\pi} \sin^2 x - (x^2 - \pi^2) dx + \int_{\pi}^{2\pi} (x^2 - \pi^2) - \sin^2 x dx \\
 &= \int_0^{\pi} \sin^2 x dx + \left[\pi^2 x - \frac{x^3}{3} \right]_0^{\pi} + \left[\frac{x^3}{3} - \pi^2 x \right]_{\pi}^{2\pi} - \int_{\pi}^{2\pi} \sin^2 x dx \\
 &= \left(\pi^3 - \frac{\pi^3}{3} \right) + \left(\frac{8\pi^3}{3} - 2\pi^3 \right) - \left(\frac{\pi^3}{3} - \pi^3 \right) = 2\pi^3 \Rightarrow \text{(P)}
 \end{aligned}$$

Section 5

Single Integer Answer Type

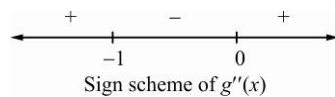
72.(3) $\therefore$ Degree of $P(x)$ is 5 with leading coefficient one.

$\therefore$ Degree of $P'(x)$ is 4 with leading coefficient five.

$$\therefore P'(x) = 5(x-1)(x-3)(x-2)^2 \quad \therefore P'(6) = 5 \times 5 \times 3 \times 4^2 = 1200$$

73.(2) Clearly, $g'(x) = f(x) = \begin{cases} (x+2)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$

$\Rightarrow g'(x)$ is a continuous function.



$\Rightarrow g'(x)$ has a local maximum at $x = -1$ and local minimum at $x = 0$.

74.(4) Circle with centre $(-6, 10)$ and radius $= \sqrt{36 + 100 - 120} = 4$

Now let $(a, 0)$ be a point on the x -axis.

If y is the distance from A to P and P to M

$$t = \sqrt{(a-2)^2 + 25} + \sqrt{(a+6)^2 + 100} - 4$$

$$\frac{dy}{da} = \frac{2(a-2)}{2\sqrt{(a-2)^2 + 25}} + \frac{2(a+6)}{2\sqrt{(a+6)^2 + 100}}$$

$\frac{dy}{dx}$ can be zero only if $a-2 > 0$ & $a+6 < 0$ not possible

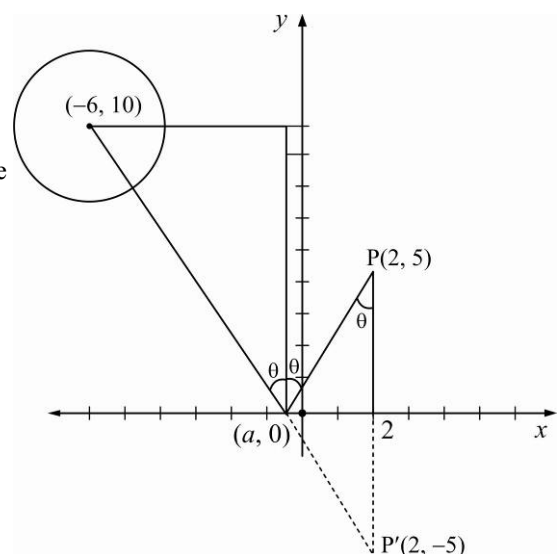
or $a-2 < 0$ and $a+6 > 0$

$$\begin{aligned}
 \text{Hence, } y_{\min} &= \sqrt{\frac{64}{9} + 25} + \sqrt{\frac{256}{9} + 100} - 4 \\
 &= \frac{17}{3} + \frac{\sqrt{1156}}{3} - 4 = \frac{17}{4} + \frac{34}{3} - 4 = 17 - 4 = 13
 \end{aligned}$$

Alternate :

$$\tan \theta = \frac{a+2}{5} = \frac{6-a}{10} \Rightarrow 10a + 20 = 30 - 5a$$

$$\Rightarrow 15a = 10 \Rightarrow a = \frac{2}{3} \quad \text{Hence, } P\left(\frac{-2}{3}, 0\right) \text{ for finding } y_{\min}.$$



75.(7) $h = p$

$$k = \frac{4}{3-p} - 1$$

$$PM = CM - PC$$

$$\text{Now, } (CM)^2 = (h-3)^2 + \frac{1}{(p-3)^2}$$

$$(CM)^2 = (p-3)^2 + \left((p-3) - \frac{4}{(p-3)} \right)^2 + 8$$

$$\therefore d_{\min} = 2\sqrt{2}$$

$$\Rightarrow PM_{\min} = 2\sqrt{2} - 1 = \sqrt{8} - \sqrt{1} \equiv \sqrt{a} - \sqrt{b} \Rightarrow (a+b) = 7$$

Alternate :

$$y = \frac{4}{3-x} - 1$$

$$y' = \frac{4}{(3-x)^2}$$

$$\therefore \frac{(3-h)^2}{4} = \frac{k+1}{h-3} \quad \text{and} \quad k+1 = \frac{4}{h-3}$$

From (i) and (ii) :

$$h=1, k=1 \text{ and } h=5, k=-3$$

$$\therefore d = 2\sqrt{2} - 1 = \sqrt{8} - \sqrt{1} \equiv \sqrt{a} - \sqrt{b}$$

$$\Rightarrow (a+b) = 7$$

76.(4) Given, $f''(x) \geq -2 \forall x \in [-3, 3]$

$$\therefore \int_{-3}^3 f''(x) dx \geq -2 \int_{-3}^3 dx \Rightarrow [f'(x)]_{-3}^3 \geq -12 \Rightarrow -12 \geq -12$$

$$\therefore f''(x) = -2 \forall x \in [-3, 3] \text{ or apply LMVT in } y = f'(x) \text{ in } [-3, 3], \text{ we get : } f''(x) = -2$$

$$\text{Now, } f'(x) = -2x + C$$

$$f'(3) = C - 6 = 0 \Rightarrow C = 6$$

$$f'(x) = 6 - 2x \Rightarrow f(x) = 6x - x^2 + C$$

$$f(0) = C = -4$$

$$\text{Hence, } f(x) = 6x - x^2 - 4$$

$$\text{Now, } g(x) = \int_0^x f(t) dt = \int_0^x (-t^2 + 6t - 4) dt$$

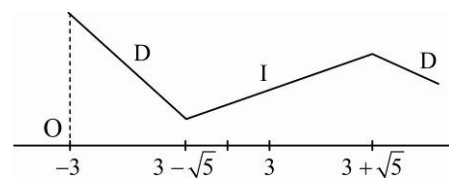
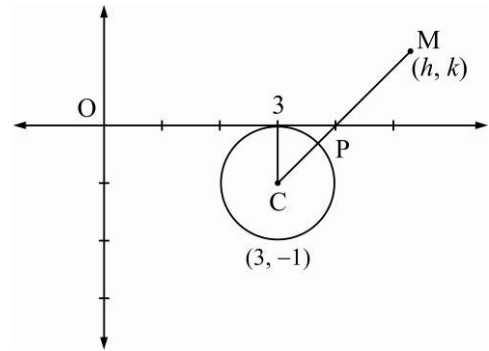
$$\therefore g(x) = \frac{-x^3}{3} + 3x^2 - 4x \text{ in } [-3, 3]$$

$$g'(x) = -x^2 + 6x - 4$$

$$g(-3) = 48 \text{ which is maximum value in } [-3, 3]$$

$$77.(3) \quad f(x) = \begin{cases} a \cos x + b \sin x & \text{if } 0 \leq x \leq \frac{\pi}{2} \\ a \cos x - b \sin x & \text{if } -\frac{\pi}{2} \leq x \leq 0 \end{cases}$$

$$\text{For } -\frac{\pi}{2} < x < 0$$



$$f'(x) = -a \sin x - b \cos x \quad \dots \text{(i)}$$

$$\text{and } f''(x) = -a \cos x + b \sin x \quad \dots \text{(ii)}$$

Since $f(x)$ has a minima at $x = -\frac{\pi}{3}$

$$\text{Now, } f'\left(-\frac{\pi}{3}\right) = +a \cdot \frac{\sqrt{3}}{2} - \frac{b}{2} = 0 = \sqrt{3}a - b = 0 \text{ f}$$

$$\text{and } f''\left(-\frac{\pi}{3}\right) = -\frac{a}{2} - b \cdot \frac{\sqrt{3}}{2} = -\frac{1}{2}[a + b\sqrt{3}] = -2a > 0$$

Hence, $a < 0$ and $b < 0$.

$$\begin{aligned} \text{Now, } I &= \int_{-\pi/2}^{\pi/2} (f(x))^2 dx = \int_{-\pi/2}^0 f^2(x) dx + \int_0^{\pi/2} f^2(x) dx \\ &= \int_{-\pi/2}^0 (a^2 \cos^2 x - 2ab \sin x \cos x + b^2 \sin^2 x) dx + \int_0^{\pi/2} (a^2 \cos^2 x + 2ab \sin x \cos x + b^2 \sin^2 x) dx \end{aligned}$$

$$\text{Hence, } I = \frac{\pi a^2}{2} + \frac{\pi b^2}{2} + 2ab = 2$$

$$2(\sqrt{3} + \pi)a^2 = 2 \Rightarrow a = -\frac{1}{\sqrt{\pi + \sqrt{3}}} \text{ and } b = -\frac{\sqrt{3}}{\sqrt{\pi + \sqrt{3}}}$$

78.(5) Let $AF = x = DE$ and $AE = y = DF$

As $\triangle CAB$ is $\triangle CED$

$$\text{So, } \frac{CE}{CA} = \frac{DE}{AB} \Rightarrow \frac{b-y}{b} = \frac{x}{c} \Rightarrow y = b\left(1 - \frac{x}{c}\right) \quad (\text{Here } BC = a, AC = b \text{ and } AB = c)$$

Now, area of parallelogram AFDE

$$= S = (AF)(EM) = xy \sin A \Rightarrow S = x \cdot b\left(1 - \frac{x}{c}\right) \sin A \quad [\text{Note : } \sin A \text{ is fixed}]$$

Now, differentiating both sides of equation (i) with respect to x , we get :

$$\frac{dS}{dx} = \frac{b}{c}(c - 2x) \sin A = 0 \Rightarrow x = \frac{c}{2}$$

$$\text{Also, } \left. \frac{d^2 S}{dx^2} \right|_{x=\frac{c}{2}} = \frac{-2b}{c} < 0$$

So, S is maximum when $x = \frac{c}{2}$

$$\text{Now, } S_{\max} = \frac{1}{4}bc \sin A = \frac{1}{2}\left(\frac{1}{2}bc \sin A\right) = \frac{1}{2}[\text{area}(\triangle ABC)] = \frac{1}{4} \begin{vmatrix} 1 & -1 & 1 \\ 4 & -2 & 1 \\ - & 3 & 1 \end{vmatrix} = \frac{20}{4} = 5 \text{ (squaring units)}$$

$$79.(2) \text{ Let } f_n(x) = \int_x^{2n} e^{-t^n} dt$$

$$f'_n(x) = 2 \cdot e^{-(2x)^n} - e^{-x^n}$$

$$\text{For maxima and minima, } f'_n(x) = 0 \Rightarrow 2e^{-(2x)^n} = e^{-x^n} \Rightarrow 2 \cdot e^{-(2^n x^n)} = e^{-x^n}$$

Taking log on both sides, we get :

$$\ln 2 - 2^n x^n = -x^n \Rightarrow \ln 2 = x^n(2^n - 1) \Rightarrow x^n = \frac{\ln 2}{2^n - 1} \Rightarrow x = \left(\frac{\ln 2}{2^n - 1}\right)^{1/n} = a_n$$

Also, $f_n''\left(x = \left(\frac{\ell n 2}{2^n - 1}\right)^{1/n}\right) < 0 \Rightarrow f_n(x)$ is maximum at $x = \left(\frac{\ell n 2}{2^n - 1}\right)^{1/n}$

Now, $\ell n a_n = \frac{\ell n\left(\frac{\ell n 2}{2^n - 1}\right)}{n} = \frac{\ell n(\ell n 2) - \ell n(2^n - 1)}{n}$

Hence, $L = \lim_{n \rightarrow \infty} \ell n(a_n) = \lim_{n \rightarrow \infty} \frac{\ell n(\ell n 2) - \ell n(2^n - 1)}{n} = \lim_{n \rightarrow \infty} \left[\frac{\ell n(\ell n 2)}{n} - \frac{\ell n\left(2^n \left(1 - \frac{1}{2^n}\right)\right)}{n} \right]$

$$L = \lim_{n \rightarrow \infty} \left(0 - \frac{n \cdot \ell n 2 + \ell n\left(1 + \frac{1}{2^n}\right)}{n} \right) = -\ell n 2$$

Hence, $e^{-L} = 2$

80.(7) $\therefore \int_a^x e^{f(t)} dt = \int_0^x g(x-t) dt + 2x + 3$

$\Rightarrow \int_a^x e^{f(t)} dt = \int_0^x g(t) dt + 2x + 3$ (Using king property)

Differential both sides, we get :

$$e^{f(x)} = g(x) + 2$$

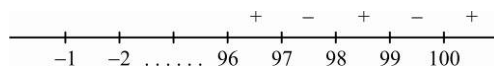
$\Rightarrow g(x) = e^{f(x)} - 2$

$\Rightarrow g'(x) = e^{f(x)} \cdot f'(x)$

$\therefore e^{f(x)}$ is always greater than zero.

$\therefore$ sign of $g'(x)$ is same as sign of $f'(x)$

$\therefore$ sign of $g'(x)$



Clearly, local extremum (maximum or minimum) will occur at $x = 99, 97, 95, \dots, 3, 1$

$\therefore$ Sum of all the values $= 1 + 3 + 5 + \dots + 99 = \frac{50}{2} [2 \times 1 + (50 - 1) \times 2] = 2500$

81.(1)

82.(1) Let $I_1 = \int_0^1 \left(1 - (1 - x^2)^{100}\right)^{200} x dx$

Put $1 - x^2 = t$ then $x dx = -\frac{1}{2} dt$

$\therefore I_1 = \int_1^0 (1 - t^{100})^{200} \left(-\frac{1}{2}\right) dt = \int_0^1 (1 - t^{100})^{200} \frac{1}{2} dt$

and similarly $I_2 = \int_0^1 (1 - t^{100})^{200} \frac{1}{2} dt = \left[(1 - t^{100})^{200} \cdot t \right]_0^1 - \int_0^1 202 \cdot 100 (1 - t^{100})^{201} (-t^{99}) dt$

(Integrating by parts)

$$= 0 - 20200 \int_0^1 (1-t^{100})^{201} - (1-t^{100}-1) dt = -20200 \int_0^1 (1-t^{100})^{202} + 20200 \int_0^1 (1-t^{100})^{201} dt$$

$$\therefore I_2 = -20200 I_2 + 20200 I_1 \quad \therefore \frac{I_1}{I_2} = \frac{20200}{20201} = \frac{p}{q} \quad \therefore (q-p)_{\text{least}} = 1$$

$$83.(2) \quad \left| 2 \left(x^2 + \frac{1}{x^2} \right) + |1-x^2| \right| = 6 - 2x^{2-1} - \frac{1}{2x^{2-1}} = \left| 2 \left(x^2 + \frac{1}{x^2} \right) + |1-x^2| \right| = 6 - \left(2x^{2-1} + \frac{1}{2x^{2-1}} \right)$$

L.H.S. ≥ 4 and R.H.S. ≤ 4

Equation will satisfy at $x = \pm 1$ $\therefore x_1 = -1$ and $x_2 = 1$

$$\text{Now, } \int_0^4 \left\{ \frac{x}{4} \right\} \left(1 + \left[\tan \left(\frac{\{x\}}{1+\{x\}} \right) \right] \right) dx$$

$$\therefore \frac{\{x\}}{1+\{x\}} = 1 - \frac{1}{1+\{x\}} \Rightarrow \frac{\{x\}}{1+\{x\}} \in \left[0, \frac{1}{2} \right)$$

$$\therefore \tan \left(\frac{\{x\}}{1+\{x\}} \right) < 1 \text{ and } \left[\tan \frac{\{x\}}{1+\{x\}} \right] = 0$$

$$\therefore \int_0^4 \left\{ \frac{x}{4} \right\} dx = \int_0^4 \frac{x}{4} dx \quad 0 \leq x \leq 4; 0 \leq \frac{x}{4} \leq 1 \quad \therefore \left\{ \frac{x}{4} \right\} = \frac{x}{4}$$

$$= \frac{1}{4} \left(\frac{x^2}{2} \right)_0^4 = \frac{1}{8} \cdot 16 = 2$$

$$84.(3) \quad Y - y = m(X - x)$$

$$\text{Now, } \left(\frac{y-0}{x-1} \right) m = -1$$

$$y \frac{dy}{dx} = 1 - x$$

$$\text{Integrating } \frac{y^2}{2} = x - \frac{x^2}{2} + C$$

$$x^2 + y^2 - 2x = C$$

This is the equation of a circle with centre (1, 0)

$\therefore 2x + 3y - 15 = 0$ is tangent at (i)

$\therefore$ Perpendicular from (1, 0) on the line = r

$$\left| \frac{2-15}{\sqrt{13}} \right| = \sqrt{1+C}$$

$$C+1=13 \Rightarrow C=12$$

Hence the curve is $x^2 + y^2 - 2x = 12$

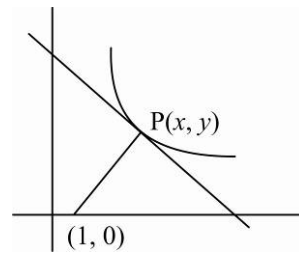
Length of tangent is $\sqrt{S_1} = \sqrt{3} \infty \sqrt{n} \quad \therefore n = 3$

$$85.(4) \quad \text{The equation of normal at } (x, y) \text{ is } Y - y = \frac{-1}{dy} (X - x) \Rightarrow (X - x) + (Y - y) \frac{dy}{dx} = 0$$

$$OA = x + y \cdot \frac{dy}{dx} \text{ and } OB = \frac{x + y \frac{dy}{dx}}{\frac{dy}{dx}}$$

$$\text{Given, } \frac{1}{OA} + \frac{1}{OB} = 1$$

$$1 + \frac{dy}{dx} = x + y \frac{dy}{dx}$$



$$(y-1)dy + (x-1)dx = 0$$

Integrating, we get : $(y-1)^2 + (x-1)^2 = C$

(3, 3) lies on this curve, $\therefore C = 8$

Hence curve is circle $(x-1)^2 + (y-1)^2 = (2\sqrt{2})^2$

The director circle of $(x-1)^2 + (y-1)^2 = (2\sqrt{2})^2$ is $(x-1)^2 + (y-1)^2 = 4^2$ $\therefore$ Radius = 4

86.(1) $y^3 y'' = -1$

$$y' y'' = \frac{-y'}{y^3}$$

Integrating both sides

$$\frac{(y')^2}{2} = \frac{1}{2y^2} + C$$

Putting $x=1, y=1$ and $y'(1)=0$

$$0 = \frac{1}{2} + C \Rightarrow C = -\frac{1}{2}$$

$$\frac{(y')^2}{2} = \frac{1}{2y^2} - \frac{1}{2}$$

$$(y')^2 = \frac{1}{y^2} - 1 \Rightarrow (y')^2 = \frac{1-y^2}{y^2} \Rightarrow y' = \pm \frac{\sqrt{1-y^2}}{y} \Rightarrow \frac{y}{\sqrt{1-y^2}} y' = \pm 1$$

Taking as +ve sign $\int \frac{y}{1-y^2} y' dx = \int 1 dx$

Putting $x=t \Rightarrow y' dx = dt$

$$\int \frac{t}{\sqrt{1-t^2}} dt = x + \lambda \Rightarrow 1-t^2 = z^2 \Rightarrow -2t dt = 2z dz$$

$$-\int \frac{z dz}{z} = x + \lambda \Rightarrow -z = x + \lambda \Rightarrow -\sqrt{1-t^2} = x + \lambda \Rightarrow -\sqrt{1-y^2} = x + \lambda$$

Putting $x=1, y(1)=1 \Rightarrow \lambda = -1 \Rightarrow -\sqrt{1-y^2} = x-1 \Rightarrow 1-y^2 = x^2 - 2x + 1$

$$y^2 = 2x - x^2$$

$$y = \sqrt{2x - x^2} = f(x) \quad [\because y \geq 0 \forall x \in D_f]$$

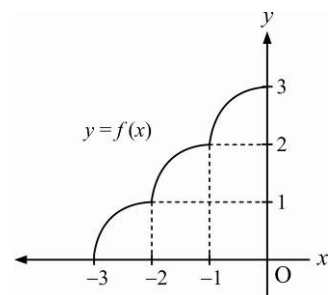
Taking as negative sign we will obtain the same function.

$$f'(x) = \frac{2-2x}{2\sqrt{2x-x^2}} = 0 \Rightarrow x=1 \text{ which is point of maxima.}$$

$\therefore$ Maxima value of $f(x) = f(1) = \sqrt{2-1} = 1$

87.(6) Area bounded by line the $y = f(x)$ and x -axis from $x = -3$ to $x = 0$, is

$$\begin{aligned} & \int_{-3}^{-2} \sqrt{x+3} dx + \int_{-2}^{-1} (1 + \sqrt{x+2}) dx + \int_{-1}^0 (2 + \sqrt{x+1}) dx \\ &= \frac{2}{3} \left((x+3)^{3/2} \right)_{-3}^{-2} + \left(x + \frac{2}{3} (x+2)^{3/2} \right)_{-2}^{-1} + \left(2x + \frac{2}{3} (x+1)^{3/2} \right)_{-1}^0 \end{aligned}$$



$$= \frac{2}{3}(1-0) + \left(-1+2+\frac{2}{3}(1-0)\right) + \left(0+2+\frac{2}{3}(1-0)\right)$$

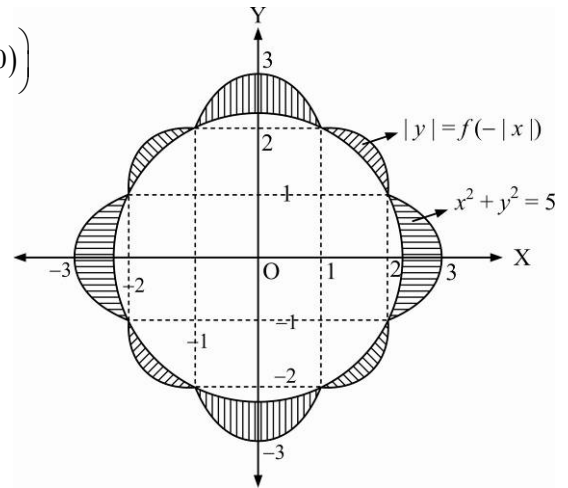
$$= \frac{2}{3} + 1 + \frac{2}{3} + 2 + \frac{2}{3} = 5$$

$$\text{Required area} = 4\left(5 - \frac{\pi}{4} \times 5\right)$$

$$= 20 - 5\pi = p + \pi - q$$

$$\Rightarrow p = 20, q = -5$$

$$\text{Hence, } p + q = 15$$



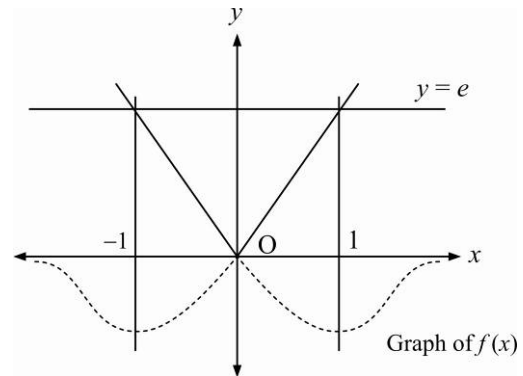
88.(3) $f(x) = \begin{cases} xe^x, & x \geq 0 \\ -xe^{-x}, & x \leq 0 \end{cases}$

For maximum area $a = e$

$$A_{\max} = 2 \int_0^1 (e - xe^x) dx$$

$$= 2 \left[ex - (xe^x - e^x) \right]_0^1 = 2(e-1)$$

$$\text{Hence, } [A] = 3$$



89.(4) $f(x) = \frac{\pi}{2} - \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \frac{\pi}{2} - 2\tan^{-1}x$

$$g(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{\pi}{2} + 2\tan^{-1}x$$

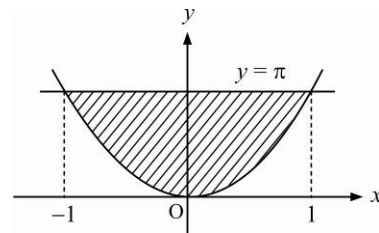
$$y = f(x) + g(x) = \pi$$

$$y = \pi x^2$$

$$\text{Required area} = 2\pi - 2 \int_0^1 \pi x^2 dx$$

$$A = 2\pi - 2\pi \cdot \frac{1}{3} = \frac{4\pi}{3} \approx 4.19$$

$$\therefore [A] = 4$$

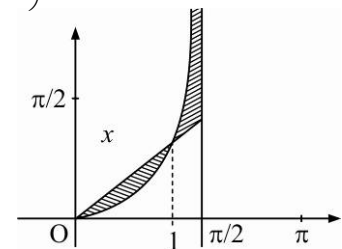


90.(1) We have $f(x) = \begin{cases} x^2 & ; x \in [0, \frac{\pi}{2}] \\ (\pi-x)^2 & ; x \in (\frac{\pi}{2}, \pi) \end{cases}$; $g(x) = \begin{cases} x & ; x \in [0, \frac{\pi}{2}] \\ (\pi-x) & ; x \in (\frac{\pi}{2}, \pi) \end{cases}$

$$\therefore \text{Required area} = 2 \left(\int_0^{\pi/2} (x-x^2) dx + \int_{\pi/2}^{\pi} (x^2-x) dx \right)$$

$$= 2 \left(\left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{\pi^3}{24} - \frac{1}{3} \right) - \left(\frac{\pi^2}{8} - \frac{1}{2} \right) \right)$$

$$= 2 \left(\frac{\pi^3}{24} - \frac{\pi^2}{8} + \frac{1}{3} \right) = \frac{\pi^3}{12} - \frac{\pi^2}{4} + \frac{2}{3} \equiv a\pi^2 + b\pi^2 + c \text{ (Given)}$$



So, $a = \frac{1}{12}, b = \frac{-1}{4}, c = \frac{2}{3}$ Hence, $(|a| + |b| + |c|) = \left| \frac{1}{12} \right| + \left| \frac{-1}{4} \right| + \left| \frac{2}{3} \right| = \frac{1}{12} + \frac{1}{4} + \frac{2}{3} = \frac{1+3+8}{12} = \frac{12}{12} = 1$